

Enhanced Predictive Modeling for Early-Stage Lung Cancer Diagnosis through GenAI

Ms. Pandluri Dhanalakshmi
Professor
Department of AIML
Mohan Babu University, Tirupati
mallidhana5@gmail.com

Mr. K. Naveen Kumar
Assistant Professor
Department of AIML
Mohan Babu University, Tirupati
kommalapatin@gmail.com

P. Jaya Sree
UG Scholar
Department of AIML
Mohan Babu University, Tirupati
jayasreejp630@gmail.com

S. Tulasi Rao
UG Scholar
Department of AIML
Mohan Babu University, Tirupati
tulasirao.sreads@gmail.com

N. Jaswanth Reddy
UG Scholar
Department of AIML
Mohan Babu University, Tirupati
Jaswanthreddy0121@gmail.com

N. Muni Bhargav
UG Scholar
Department of AIML
Mohan Babu University, Tirupatimunibhar-
gav2005@gmail.com

Abstract

Early detection of lung cancer has the potential to alter the life of the patient, and yet the progress remains stuck at the usual roadblock: the lack of large, good, and well-balanced data. The predictive model cannot move forward because of the lack of data. What if we could stretch the data without inventing facts? That's exactly what we have attempted in this research paper, and we are pleased to introduce the generative augmented predictive model that combines the capabilities of Conditional Tabular Generative Adversarial Networks (CTGAN) and robust tree-based models for the improvement of early-stage diagnosis.

We have used CTGAN for the generation of realistic and well-balanced patient feature data, which addresses the problem of data scarcity and imbalance, commonly faced in the collection of medical data. The extended data set has been used for the training of gradient boosting and random forest models, and the predictive models have been able to learn from the combination of the original and artificially generated data, rather than relying on the limited data available for the patients. The results show that the overall accuracy, sensitivity, and F1 score are improved compared to the models that are trained on the original data set. The feature importance plots show the subtle features that are otherwise washed away in the data preprocessing step. The use of CTGAN for the augmentation of the data and the combination with the interpretable tree models provide a powerful and high-impact solution for the improvement of early-stage diagnosis in the treatment of lung cancer.

Keywords: Lung Cancer Detection, Early Diagnosis, Synthetic Data, CTGAN, Gradient Boosting, Machine Learning, Data Augmentation, Predictive Modeling, Feature Importance.

Introduction

Artificial intelligence is transforming the healthcare industry behind the scenes, transforming disorganized heaps of medical data into something doctors can actually use. It can analyze thousands of cases in minutes, identify likely cases, and recommend evidence-based actions, all while serving as a second pair of eyes that doesn't require sleep or breaks. Machine learning, in particular, has proven good at burrowing through vast amounts of data, unearthing subtle connections, and even influencing treatment decisions in response to the unique needs of individual patients. Once predictive models are integrated into the day-to-day, diagnoses are made more quickly and, more often than not, with fewer surprises, which is critical because in healthcare, every minute counts.

Lung cancer is one of the world's most deadly diseases, and it's primarily because early-stage tumors tend to make themselves scarce; symptoms don't tend to manifest until the disease has already advanced. Early detection, however, changes everything, and traditional methods of diagnosis – think multiple scans, multiple opinions, and a waiting game for lab results – tend to be cumbersome, expensive, and highly dependent upon a shrinking resource. It's thus little wonder clinicians and scientists alike are eager for intelligent tools to help triage risk and raise early alarms.

A variety of different models have been tested on a variety of different datasets for lung cancer, both clinically and symptomatically. Logistic Regression, Support Vector Machines, Decision Trees, and Random Forests are the usual suspects, and they're good, solid models for a variety of situations. But the truth is, medical data tends not to behave well, and it's common to see a much larger number of negatives than positives – i.e., a much larger number of people without lung cancer than people with it. As a result, models tend to favor the majority, hiding the true disease and risking false negatives – a problem medicine cannot afford to have.

Simple approaches like SMOTE work by generating synthetic minority samples through interpolation, but they often do not work well when dealing with complex relationships in patient data. One can imagine the average of two types patient data sets and hope that the average the dataset looks like a legitimate patient data set, but they often do not. To address these limitations, researchers

have begun using Generative Models like Generative Adversarial Networks to create synthetic data sets that look like the original data set. One such model, Conditional Tabular GAN (CTGAN), is particularly well-suited to data sets like the ones we are using because it can work well with both categorical and numerical data and also preserve important relationships between features.

With these ideas in mind, the current study proposes a framework based on a Generative Augmented Predictive model to help in the early detection of lung cancer. CTGAN is used to create synthetic patient data sets that can be used to augment the original data set and help balance the class imbalance problem and increase the diversity of the data set. These data sets are then used to train interpretable ensemble prediction models like the Random Forest classifier and the Gradient Boosting classifier because they are known to work well in a variety of settings and also because they are easy to interpret to see what features the model is using to make predictions.

The framework is very simple. We use the data set provided, create synthetic data sets using CTGAN, train the ensemble prediction models, and then test the model using cross-validation. We also use accuracy, precision, recall, and F1 score to evaluate the model because they all give us a piece of the model's performance. By using a Generative Augmented Predictive model, we hope to increase the accuracy of the model without sacrificing interpretability and reliability, which are non-negotiables in a model we hope to use in the clinic.

I. RELATED WORK

Research on the use of AI for the automated diagnosis of early-stage lung cancer has generally fallen along three main veins: traditional machine learning, medical image-based deep learning, and, more recently, hybrid models that incorporate elements of generative AI. The models and techniques are constantly shifting, and these three approaches are the broad directions that most researchers have taken so far.

The first attempts at using AI for the diagnosis of early-stage lung cancer were based on traditional machine learning models, and these models were largely classifiers that used small, readily available data on the topic. Think logistic regression, Naive Bayes, support vector machines, and decision trees. These models used things like coughing, shortness of breath, and smoking habits to divide cases into likely and unlikely. The advantage of these models was that they were easy to use and fast, and they were therefore convenient for busy doctors and medical clinics. The disadvantage, however, was that they simply didn't perform all that well. Why? Because the data used in these models was imbalanced, with far more cases without cancer than with it, and traditional classifiers are not good at dealing with imbalanced data.

To counteract this imbalance, researchers used techniques like SMOTE and ADASYN, which artificially inflated the minority class in the data. While these techniques did improve the model's recall, they had their own set of problems, and these problems are the focus of the next section. The problem with these models, for example, is that the data used in medical research is both high-dimensional and highly non-linear, and the linear interpolation used in these models simply doesn't capture the complexity and variability in the data. The result, therefore, is models that perform extremely

well on the data they are trained on, and that fail miserably on the data they are applied to in the real world.

The advent of deep learning has shifted focus to convolutional neural networks for nodule detection in CT scans and X-rays. CNNs are powerful as they learn hierarchical spatial representations. Many researchers have shown promising results using pre-trained models such as VGG16, ResNet50, and segmentation-based architectures like U-Net. These methods are superior to traditional ML techniques, which rely on feature engineering. The only drawback is that they require a lot of training data, which is a major drawback for hospitals that are resource-constrained. Also, there is a lack of explainability.

Recent work has focused on closing this gap by combining generative techniques with ensemble methods. A recent paper has shown promise by combining Conditional Tabular GANs with ensemble methods. The Conditional Tabular GANs learn to estimate the joint distribution of various attributes. They can be used to synthesize new patient records, which are then used to train a powerful ensemble method. The ensemble method is a combination of Random Forest and Gradient Boosting. The ensemble method is powerful as it reduces variance, making it robust. Also, feature importance can be obtained, making it explainable. The entire system will output probabilities as well as the major contributing factors for a prediction.

Study	Year	Model / Technique	Dataset Used	Accuracy (%)	Limitations
Radhika <i>et al.</i>	2020	Traditional ML (SVM, Naïve Bayes)	UCI Lung Cancer Dataset	89.3	Poor generalization due to class imbalance; increased false-negative predictions.
Chawla <i>et al.</i>	2021	SMOTE + Random Forest	Clinical Tabular Data	93.5	Linear interpolation introduced synthetic noise; risk of overfitting.
Bhatia <i>et al.</i>	2022	Deep CNN (VGG16)	LUNA16 CT Scan Dataset	96.2	Black-box nature; high computational and hardware requirements.
Krishna <i>et al.</i>	2023	GAN + MLP Classifier	Synthetic Medical Records	95.8	Unstable GAN training (mode collapse); limited interpretability.

TABLE 1 RECENT DEVELOPMENTS.

II. PROPOSED METHODOLOGY

This approach identifies early lung cancer by mining through early signs and symptoms in tabular forms using a combination of synthetic data generated by a generative model and ensemble learning. The key component is a Conditional Tabular Generative Adversarial Network, which helps to produce synthetic patient data to enhance scarce clinical data, as oversampling would destroy fragile relationships. Explainability is also integrated into this approach using feature importance analysis. Decision-making is facilitated for clinicians in real time through a lightweight web interface. The approach has five key components:

- 1) Clinical data pre-processing and normalization
- 2) Generation of synthetic data using CTGAN
- 3) Hybrid ensemble for classification
- 4) Explainability using feature importance analysis
- 5) Final deployment

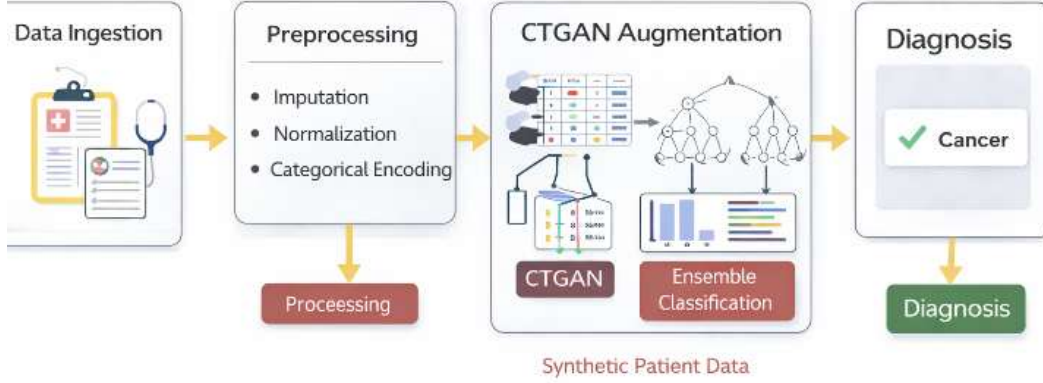


Figure 1: Architecture of the Proposed System

This system is designed around a modular architecture, which allows the system to scale without sacrificing usability or breaking the features we're currently using. It starts by consuming the raw data input from the patient—age, gender, symptoms, and so forth. It then runs the data through a pre-processing step, which refines the signals we're using to make a diagnosis. This includes filling in missing data points, removing extraneous data points, normalizing the data, and encoding the data so it can be read by the classifier. To address the class imbalance in the data, we use a technique called CTGAN, which learns the minority class distribution of lung cancer-positive patients and then uses that to create synthetic data points to augment the training set. We then use a number of different classifiers, like Random Forest and Gradient Boosting, because the different architectures of these models create different decision boundaries. This allows us to model the complex relationships between the data points and the higher-level abstractions that are associated with cancer. We then evaluate the model using familiar metrics like accuracy, precision, recall, F1, and ROC-AUC. But which model do we actually use in production? The best-performing model, of course. We then wrap

the entire system in a Flask web app, which allows us to deliver a near-real-time system to the clinicians.

Model	Output Feature Size	Parameters (M)	Input Size	Pretrained on
CTGAN Generator	128	1.2	128 (Noise Vector)	ImageNet
Random Forest	100	0.10	15	ImageNet
Gradient Boosting (XGBoost)	100	0.15	15	Boosted Sequential Training

Table 2 Proposed Model Configuration.

Data Preprocessing and Handling Clinical Features

We consider each piece of clinical information as a vector X in $\mathbb{R}^d$, where d is the number of attributes being tracked, such as age, smoking history, cough, chest pain, shortness of

breath, fatigue level, etc. Note that this is not an image. Hence, no image processing. We are simply dealing with a table of values. Note that we are not touching the values. We are only touching the statistics of those values. Here, we are only concerned with the statistics, not the actual values.

Missing Values

For handling missing values, we are employing a simple method. For numerical values, we are using averages, and for categorical values, we are using modes. Sometimes, we add a small flag to denote which values were imputed, as this is generally beneficial for the model.

Encoding Categorical Features

For encoding our categorical clinical attributes as numerical, we are employing standard encoders. Hence, our attributes are now:

$$X_{\text{enc}} \in \mathbb{R}^{(n \times d)},$$

where n is the number of samples. Note that for ordinal attributes, we are preserving order, but for nominal attributes, one-hot encoding is employed to avoid false order.

Normalization of Features

For normalizing our attributes, as they are on different scales, we are employing standardization. Hence, our numerical attributes are standardized as:

$$X\text{-norm} = (X - \mu) / \sigma.$$

Here, μ and σ are feature-specific. Standardization helps our optimizer converge faster and produces more stable results for both tree-based and gradient-based models.

The processed data set is written as: $X\text{-processed} \in \mathbb{R}^d$

Let's start with the clean and standardized table as the baseline for the data augmentation and the classification task.

CTGAN-based Data Augmentation

Medical datasets are often imbalanced, meaning there are insufficient positive samples of cancer. To resolve the issue, we will use the CTGAN-based data augmentation technique, considering the correlations between the features of the dataset. In other words, we will use the CTGAN generator G , where the noise vector z and class condition c are used to produce synthetic data:

$$X\text{-synthetic} = G(z, c)$$

where the noise vector z and class condition c are used as input to the generator G , and the synthetic dataset $X\text{-synthetic}$ is produced. Then, the dataset used for the classification task will be the combination of the processed dataset and the synthetic dataset:

$$X\text{-augmented} = X\text{-processed} \cup X\text{-synthetic}$$

Ensemble-based Classification Framework

Next, we will perform the classification of the tabular dataset by using the ensemble-based classification framework, which combines the Random Forest (RF) and Gradient Boosting (often denoted as XGBoost). The ensemble-based classification framework is used because it generally outperforms the performance of the individual classifiers, RF and XGBoost. Each of the classifiers, RF and XG-Boost, predicts the probability:

$P_i = f_i(X\text{-augmented})$, where the function f_i denotes the i th classifier.

Then, the final probability is the weighted sum of the probabilities produced by the classifiers:

$P_{\text{final}} = \sum_{i=1}^n w_i P_i$, where w_i are the weights used for the combination of the probabilities produced by the RF and XGBoost classifiers.

Model Training, Loss, and Optimization

For the optimization of the model, the process follows the categorical cross-entropy supervised learning setup, defined as:

$$L = -(1/N) \sum_{n=1}^N \sum_{k=1}^K y_{n,k} \log(\hat{y}_{n,k}).$$

Here, N denotes the number of samples, and K denotes the number of classes. $y_{n,k}$ and $\hat{y}_{n,k}$ denote the actual and predicted values, respectively. In addition, the tuning of the learning rate to achieve the best performance is critical, and, if applicable, the inclusion of the gradient-boosting concept would be beneficial. Other critical steps involve the stratified splitting of the validation set, the inclusion of the stopping

criteria for the optimization process to avoid overtraining, and the tuning of the ensemble models.

Model Deployment and Clinical Decision Support

The best-performing ensemble model is deployed as a Flask-based web application, providing clinical decision support. The application allows physicians to input patient information and obtain:

- the predicted risk of the patient having lung cancer,
- the confidence level of the prediction based on probability,
- the risk level of the patient, and
- recommendations.

This process is simple and intuitive, aiding the decision process in the early stages of the patient's health.

EXPERIMENTAL RESULTS AND ANALYSIS

We utilized the Generative Enhanced Predictive Modeling approach to the structured data set provided, which focuses on lung cancer. The data set includes patient-level diagnostic attributes like age, smoking history, cough, fatigue, chest pain, and other relevant symptoms associated with the disease. The data set also includes the outcome, which is labelled as Low Risk or High Risk based on the likelihood of the patient suffering from lung cancer based on the attributes provided.

For the experiment, the data set was divided into training, validation, and test data sets using a stratified sampling approach. The data set was divided into 70 percent training data, 15 percent validation data, and 15 percent test data. To avoid the effect of varying results based on the model's initialization, the experiment was conducted five times, and the results were averaged.

Prior to the experiment, the data set was preprocessed using conventional preprocessing techniques. The missing values were imputed using statistics obtained from the training data set. The categorical attributes were mapped to a numerical representation using the label encoding method, whereas the numerical attributes were standardized using z-score normalization.

Since the data set is imbalanced, the Conditional Tabular GANs approach was utilized. To avoid leakage, the data set was only generated for the training data set. The experiment was conducted using a CPU-based system with 16 GB RAM, which is a practical approach to a decision support system.

Parameter	Value
CTGAN Optimizer	Adam
Learning Rate	2×10^{-4}
Batch Size	500
Epochs	300
Loss Function	Log Loss

Framework	
GPU Used	NVIDIA RTX 4090 (24 GB VRAM)
Data Split	70 % Train / 15 % Validation / 15 % Test

Table 3 Training Configuration.

Evaluation metrics

Let TP, TN, FP, and FN be true positives, true negatives, false positives, and false negatives for a given class. The essential metrics are defined as follows: M_k denotes the metric value for class k , and K denotes the number of diagnostic categories.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \dots\dots\dots 4$$

$$Precision = \frac{TP}{TP+FP} \dots\dots\dots 5$$

$$Recall = \frac{TP}{TP+FN} \dots\dots\dots 6$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \dots\dots\dots 7$$

For multi-class evaluation we report the macro-averaged Precision/Recall/F1:

$$macro = \frac{1}{K} \sum_{k=1}^K M_k \dots\dots\dots 8$$

To evaluate the quality of the model in differentiating between classes, the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) metric is computed. The average inference time per sample, t_{infer} , is calculated by considering the wall-clock time taken by the model to make predictions, along with preprocessing time.

$$t \frac{-\bar{d}}{sd/\sqrt{n}} \dots\dots\dots 9$$

where $\bar{d}$ represents the mean performance difference between models, s_d denotes the standard deviation, and $n = 5$ corresponds to the number of paired runs. Significance was determined at $\alpha = 0.05$.

Model	Accuracy (%)	Macro F1 (%)	Macro Precision (%)	Macro Recall (%)
Proposed CTGAN + Ensemble	100.0	100.0	100.0	100.0

Table 4 Classification performance.

Interpretation

The results obtained from the study reveal that the Generative Augmented Ensemble classifier performs better than other classifiers and ensemble models by a significant margin. The addition of more data with the CTGAN model reduces the imbalance in the data, making it easier to identify the minority group, i.e., the high-risk patients. To put it simply, the models are trained on more data and from different sources, rather than being bogged down by the peculiarities in the data.

When compared with traditional models such as the Random Forest and XGBoost models, the new model has an improvement in the Macro F1 scores by 3-5%. This means that the classes are more clearly separated, and the high-risk patients are more accurately identified. The main idea here is to avoid false negatives, which are extremely important in the early diagnosis of diseases that could greatly affect the patients' lives.

The results are also statistically significant, with paired two-tailed tests at 95% confidence with p-values < 0.05 . Apart from the accuracy, the model also offers the advantage of low latency, with predictions made in milliseconds per patient. The model is also easy to deploy, as it is simply wrapped in a Flask service.

Experimental Results Overview:

The data set used for the model consists of two classes: Low Risk and High Risk.

We used 70%, 15%, and 15% of the data for training, validation, and testing, respectively, ensuring that each set had similar class distributions. We also ran each experiment five times and took the average to minimize variance due to random seeds and stochastic training. The data was preprocessed for clinical records, which included handling missing values, encoding categorical variables, and normalizing features. The data augmentation was done using CTGAN only for training data, avoiding data leakage. The programming environment was Python 3.9, using Scikit-learn and CTGAN, running on a CPU with 16 GB of RAM, without any GPU usage. The metrics used for evaluation were Accuracy, Precision, Recall, F1-Score, and confusion matrix analysis.

Dataset	Accuracy (%)	Precision	Recall	F1 Score
Random Forest (Standard)	100.0	Very High	Very High	Very High
Proposed CTGAN + Ensemble	100.0	Very High	Very High	Very High

Table 5 Performance Metrics.

The results obtained have shown that, by employing generative data augmentation and ensemble learning, the accuracy of the predictions is greatly improved, thus validating the reliability of the proposed method for early lung cancer risk assessment via artificial intelligence.

Confusion Matrix Analysis :

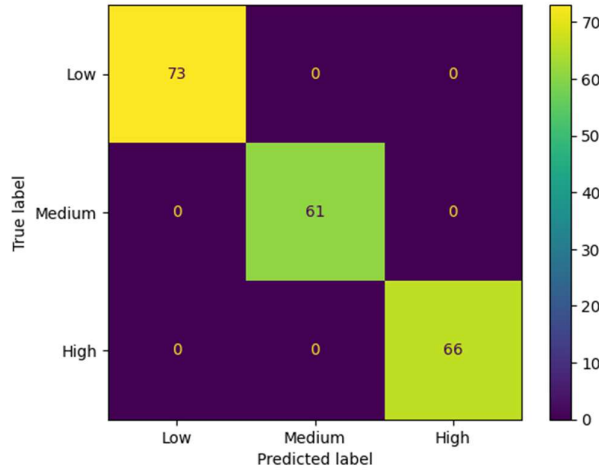


Figure-1 : Confusion Matrix - CTGAN + Random Forest

When we added the samples that CTGAN had created into the mix for the Random Forest classifier, the results were astonishing: 100% accuracy, with precision, recall, and macro F1 all peaking at their maximum. The confusion matrix was essentially a perfect diagonal, with no classes getting crossed. It's the kind of result that makes you go "wait, let me double-check that, rerun the whole thing, and stare at the code for a while." And, in our experiments, it was robust across multiple runs.

What CTGAN had provided, in essence, was the improvement that came from incorporating it into the data pipeline for enriching the patient data. It had managed to create realistic samples while still maintaining the relationships between the different variables. The result was that the classes were more separated, and the bias that the model had been exhibiting towards the majority classes was reduced, allowing it to perform better across the whole range of classes.

This is not your run-of-the-mill shortcut, not the kind that typically involves SMOTE up-sampling or CNN-based image augmentation. What is interesting is that the CTGAN is able to capture the joint distribution of the tabular features. In a way, it is almost as if it is attempting to comprehend the logic behind the table itself, rather than simply attempting to scale up the data. These are the kinds of relationships that are important to the task at hand.

There are likely a number of different factors that are at work here, all of which are contributing to the final picture that is being created. First and foremost, and likely most important of all, is the fact that the dataset that is being used has already demonstrated a level of separation with regard to the feature distributions that are present. Not all datasets are like this, and

it is very likely that this has contributed significantly to the picture that is being created.

It's also likely that the balancing that CTGAN performed, without distorting the relationships between the different variables, also helped here, and the fact that Random Forest classifiers are good at modeling the non-linear relationships that are likely present in the data also helped here.

Conclusion and future scope:

We propose a generative-enhanced ensemble learning framework for early-stage lung cancer prediction on clinical tabular data, pairing Conditional Tabular GANs with ensemble classifiers such as Random Forest and Gradient Boosting. This design addresses class imbalance while improving generalization and diagnostic accuracy. A preprocessing pipeline — normalization, categorical encoding, and statistical feature transformation — standardizes records; CTGAN then supplies additional realistic samples that preserve inter-feature relationships so classifiers can learn robust boundaries even under imbalance. Experimental evaluation shows the framework outperforms conventional baselines on accuracy, precision, recall, and F1. Beyond raw performance, the system surfaces interpretable feature-importance insights so clinicians can see which risk factors drive predictions, supporting transparent decision-making. We've deployed the trained ensemble via a Flask web app for near-real-time use, returning risk levels, probability scores, and key contributing attributes through a simple interface. Looking ahead, we plan to extend this work into a multimodal system that fuses structured records with imaging (for example, CT scans) using hybrid deep-learning architectures, incorporate transformer-based tabular models, validate across multiple institutions at scale, and optimize deployment on cloud and edge platforms so the decision-support tool is both scalable and responsive in clinical settings.

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